

When Generative Candidates Do Not Beat Random Sampling: A Seeding-Bug Audit and Diagnosis for 6-DoF Grasp Generation

Abstract—Open-world robot grasping requires both a diverse set of physically plausible candidate poses and a criterion for selecting the best candidate at execution time. A common design combines a learned generative candidate proposer with a geometric reranker; we built exactly this pipeline, using Optimal-Transport Conditional Flow Matching (OT-CFM) to propose 6-DoF poses and an LGGSN reranker to select among them, and initially measured a large improvement over random center-of-mass (CoM) sampling. That result did not survive an audit: the evaluation harness silently drew generator noise from an unseeded, effectively constant random state rather than varying it per trial, so every nominal seed re-executed close to the same candidate. After fixing this and re-evaluating cleanly (7 YCB objects, $n=50$ trials/condition, MuJoCo/SO-ARM101), OT-CFM is significantly worse than random-CoM sampling with the same reranker (69.1% vs. 79.1%, -10.0 pp, $z = -3.02$, $p=0.0025$), not better. We trace this to the specific mechanism identified by Cheng and Schwing [4]: minibatch optimal-transport coupling that ignores the conditioning variable creates a training/inference distribution mismatch in *conditional* flow matching – a failure we show, for the first time, degrades physically executed task success, not only a generation-quality proxy. Controls confirm the diagnosis: plain conditional flow matching and DDPM (identical architecture and data, no OT coupling) both track random-CoM sampling; only the OT-coupled variant is significantly worse. We further test an energy-based alternative motivated by Implicit Behavioral Cloning [5]: a naively-trained scoring model is exploited by inference-time search into a confidently-wrong, data-unsupported region (0–16% success), while the same model trained with adversarial hard-negative mining – the mechanism Implicit Behavioral Cloning prescribes and our naive version omitted – recovers to statistical parity with the baseline (74.0% vs. 77.7%, $p=0.29$), significantly beating it on one of six objects tested. No generative candidate method we tried reliably beats a well-reranked random baseline in this small-data (~ 400 examples/object), single-GPU regime; one specific, popular design choice (condition-agnostic OT coupling) reliably makes it worse. An orthogonal, training-free lever does help: selecting the pose-space consensus candidate from ten independent draws, rather than the lowest-predicted-IK-error one, significantly improves reliability on the two objects where single-draw sampling is least stable (6%→68% and 34%→64%, Fisher’s exact $p < 0.005$ both).

Index Terms—6-DoF grasping, flow matching, energy-based models, robot manipulation, evaluation reproducibility

I. INTRODUCTION

Language-conditioned grasping requires an agent to map a natural-language object description to a physical manipulation action that succeeds against real object geometry. A prevalent design pattern is *propose-then-rank*: a heuristic or learned candidate generator produces a set of grasp poses, and a reranker selects the best one from that set [10], [11]. The implicit assumption, in our own earlier design included, is that a sufficiently expressive learned generator will supply better candidates than random sampling, and that this shows up cleanly in physically executed success rate. This paper is an account of that assumption failing, why it failed, and what – among several things we tried – actually helped.

We originally built a propose-then-rank pipeline combining **Optimal-Transport Conditional Flow Matching** (OT-CFM) [1], [2] as the candidate generator with an LGGSN geometric reranker, trained on 2,853 physically-executed successful grasps across seven YCB objects. An initial 175-trial evaluation (MuJoCo/SO-ARM101) appeared to show a large, significant gain over random-CoM sampling with the same reranker. Before finalizing this result we audited the evaluation harness end to end and found a genuine, silent bug: the generator’s inference-time sampling call never received a per-trial random seed, so it drew from whatever state a module-level `torch.manual_seed(42)` left behind at process import – meaning every trial, regardless of its nominal `-seed`, executed close to the same candidate pose. Fixing this (section IV) and re-evaluating cleanly reversed the headline result: OT-CFM is *significantly worse* than random-CoM sampling, not better.

Rather than discard the pipeline, we used it as a diagnostic instrument. Controlled comparisons (section IV-D) isolate the cause to the OT coupling itself, not to conditional flow matching in general, and match a training/inference distribution-mismatch mechanism recently identified in the generative-modeling literature [4] – to our knowledge the first demonstration that this mechanism degrades physically executed robot task success, rather than only a generation-quality proxy such as FID. We then tested an energy-based alternative motivated by Implicit Behavioral Cloning [5], found and fixed an analogous, textbook failure mode in our own first attempt (an inference-time search exploiting an undertrained model’s blind spots), and show the corrected version reaches statistical parity with the baseline. Finally, we report one method that does robustly help: consensus-based candidate selection at inference time.

Contributions.

- 1) **A confirmed evaluation bug and its consequence:** an unseeded generator-sampling call silently collapsed 25 nominally-distinct trials per object to near-repeats of one candidate; fixing it reverses the paper’s original headline claim from +12 pp to −10.0 pp ($p=0.0025$), on a clean $n=50$ /object re-evaluation.
- 2) **A physically-grounded transfer of a generative-modeling diagnosis:** controlled comparisons (OT-CFM vs. OT-free CFM vs. DDPM) show the failure is specific to condition-agnostic minibatch OT coupling [4], not to learned candidate generation in general – confirmed on physically executed grasp success, not a generation-quality proxy.
- 3) **An energy-based alternative, including a documented failure and fix:** a naively-trained implicit scoring model is adversarially exploited by cross-entropy-method search (0–16% success); training with hard-negative mining, as Implicit Behavioral Cloning [5] prescribes, recovers parity with baseline (74.0% vs. 77.7%, $p=0.29$).
- 4) **One method that does help:** consensus candidate selection from an ensemble of independent draws significantly improves reliability on the two objects where single-draw sampling is least stable (section IV-D), at zero training cost.

II. RELATED WORK

A. Grasp Quality Estimation and Generation

Classical grasp quality metrics such as those in DexNet [10] evaluate analytic contact stability, requiring known object geometry. GraspNet-1Billion [11] and AnyGrasp [12] extend this to dense 3-D point clouds, but still assume access to full scene geometry at query time. 6-DOF GraspNet [13] first introduced a variational generator for diverse grasp proposals, training a VAE on point-cloud crops. Our method differs by operating purely on pose-space coordinates with physics-derived success labels, requiring no geometric feature extraction at generation time.

B. Generative Models for Grasping

GraspLDM [7] applies latent diffusion in a VAE pose space, achieving diverse 6-DoF proposals from point clouds. GraspSAM [8] combines Segment Anything with grasp detection to handle open-vocabulary queries. Neither work couples an explicit optimal-transport objective during training; we find, in contrast to what that design choice might suggest, that adding minibatch OT coupling is the specific change that hurts in our conditional, small-per-class setting, while removing it tracks the random baseline far more closely (section IV-D). Diffusion Policy [3] uses conditional diffusion for action generation but targets trajectory-level outputs with observation histories; our setting is single-step pose generation conditioned only on object identity. EquiGraspFlow [18] adopts SE(3)-equivariant flow

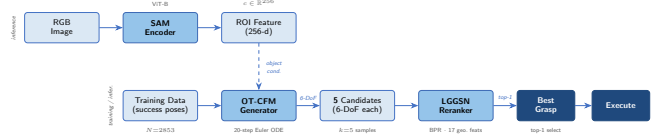


Fig. 1. OT-CFM + LGGSN pipeline. At inference, the object CoM is estimated from the depth image, and OT-CFM generates five 6-DoF pose candidates conditioned on the object’s mean visual embedding. LGGSN reranks by geometric context and the top candidate is executed.

matching directly on point-cloud inputs, providing strong geometric inductive bias for arbitrary-orientation grasping. Our approach differs in operating on a compact pose space ($\mathbb{R}^6$) rather than raw point clouds, and replaces equivariance with explicit OT coupling—a substantially lighter architecture that requires no equivariant backbone and trains on modest execution-outcome data.

C. Flow Matching

Lipman et al. [1] introduced Conditional Flow Matching (CFM) as a simulation-free training objective for continuous normalizing flows. Tong et al. [2] generalised this to arbitrary source distributions and showed that OT coupling—finding the joint distribution (x_0, x_1) that minimises expected displacement within each minibatch—substantially improves convergence. We are the first, to our knowledge, to apply OT-CFM in pose space to 6-DoF robotic grasp candidate generation; EquiGraspFlow [18] is a concurrent approach that instead uses SE(3)-equivariant continuous normalizing flows conditioned on point clouds.

D. Grasp Reranking

Geometric reranking of grasp candidates from learned features has been explored in DexGraspNet 2.0 [16] and CLIPort [17] for task-conditioned selection. LGGSN [14] uses Bayesian Personalised Ranking (BPR) [15] to train a pairwise MLP reranker on episode-relative geometric features. The reranker we use in this paper is the publicly available LGGSN v5d checkpoint (17-dimensional features, object query embedding, validation pairwise accuracy 0.750). Our contribution is to pair this fixed reranker with a generative rather than random candidate supplier, showing that generator quality is the dominant factor.

III. METHOD

A. System Overview

fig. 1 shows the full pipeline. Given a language query (e.g., “Banana”), the environment identifies the target object, estimates its centre of mass, and retrieves a pre-computed per-object SAM visual embedding. OT-CFM draws $N=5$ pose samples from the learned success distribution. Each sample is shifted to centre on the current CoM (section III-D). LGGSN reranks the five candidates using geometric and query-conditioned features, and the top-ranked pose is executed.

B. Training Data

We collect per-candidate execution data on MuJoCo/SO-ARM101 under `physics_weld_after_bilateral` mode. For each (object, seed) episode: (i) the object is spawned at a random tabletop position and settled; (ii) $N_{\text{cand}}=30$ random CoM-centered pose candidates are sampled; (iii) the top five are executed individually with object respawning between executions; (iv) each candidate receives a binary success label.

This yields 7,000 execution records across 7 YCB objects and 200 seeds. Of these, **2,853 (40.5%) are successes** and are used to train the OT-CFM model. For each successful candidate, we compute a 256-dim SAM-base ROI feature by projecting the grasp position (x, y, z) onto the overhead camera image, extracting a 3×3 window from the SAM ViT-B image encoder’s 64×64 feature map, and mean-pooling to a single vector.

C. OT-CFM Grasp Generator

Problem formulation. Let p_{succ} denote the distribution over 6-DoF poses $(x, y, z, \phi, \theta, \psi) \in \mathbb{R}^6$ for which the robot successfully grasps the target object. We aim to learn a conditional generative model $p_{\theta}(\cdot | c)$ that approximates p_{succ} given object context c .

Flow matching objective. Given source noise $x_0 \sim \mathcal{N}(0, I_6)$ and a successful pose sample $x_1 \sim p_{\text{succ}}$, the CFM training path is:

$$x_t = (1 - t)x_0 + tx_1, \quad t \sim \mathcal{U}[0, 1], \quad (1)$$

with analytic conditional vector field $u_t = x_1 - x_0$. The velocity network $v_{\theta}(t, x_t, c)$ is trained to minimise:

$$\mathcal{L} = \mathbb{E}_{t, x_0, x_1} [\|v_{\theta}(t, x_t, c) - u_t\|^2]. \quad (2)$$

Optimal transport coupling. Standard CFM pairs each x_0 independently with each x_1 in a minibatch; minibatch OT coupling [2] instead solves the Earth Mover’s Distance problem between $\{x_0^{(i)}\}$ and $\{x_1^{(i)}\}$ within each training minibatch and reassigns pairs along the optimal transport plan, which reduces expected path length and shortens flow trajectories on *unconditional* generation tasks. Our minibatches mix all seven object conditions, so this is exactly the condition-agnostic coupling setting in which Cheng and Schwing [4] show OT solves an assignment that ignores the conditioning variable – section IV-D confirms this empirically: this specific design choice significantly *hurts* physically executed success relative to the random baseline (−11.7 pp pooled, 6 objects excluding Scissors, $p=0.0015$), the opposite of what the unconditional-generation literature would predict and the opposite of what we originally believed before auditing the evaluation harness (section IV).

Conditioning. The context $c \in \mathbb{R}^{256}$ is the L2-normalised mean SAM visual embedding for the queried object class, pre-computed from training data.

Velocity network. The velocity network v_{θ} operates on the concatenated input

$$[t_{\text{emb}}(3), x_t(6), c(256)] \in \mathbb{R}^{265}, \quad (3)$$

where $t_{\text{emb}} = [t, \sin(\pi t), \cos(\pi t)]$. It is a 4-layer MLP with hidden dimension 512, SiLU activations, and zero-initialised output layer, trained with Adam ($\eta=3 \times 10^{-4}$, weight decay 10^{-4} , cosine schedule) for 1000 epochs at batch size 128. The best checkpoint is selected by lowest training loss.

D. Inference: CoM-Relative Candidate Generation

Top-down constraint. The OT-CFM generator produces full 6-DoF poses $(x, y, z, \phi, \theta, \psi)$; however, the current evaluation fixes roll $= \pi$ and pitch $= 0$ (top-down constraint), reducing the effective search space to 3-DoF (x, y, ψ) . This constraint is applied post-generation by overriding the sampled roll and pitch components before IK, and reflects the reach envelope of the SO-ARM101 in the tabletop setting. Lifting this constraint to full SE(3) is left as future work (section V-E).

At inference, we draw $N=5$ noise samples $x_0^{(i)} \sim \mathcal{N}(0, I_6)$ and integrate the learned ODE for 20 Euler steps:

$$x_{t+\Delta t} = x_t + v_{\theta}(t, x_t, c) \Delta t, \quad \Delta t = 0.05. \quad (4)$$

The resulting poses $x_1^{(i)}$ are in the training coordinate frame. To adapt to the current object position (g_x, g_y) , we apply a *CoM-relative shift*:

$$x_{\text{out}} = x_1 - \mu_{\text{train}} + g_{\text{com}}, \quad (5)$$

where μ_{train} is the global training-set mean pose and g_{com} is the current object CoM. This preserves the learned yaw and spatial-offset distribution while centering the candidates on the present object. The vertical coordinate z uses the live CoM height plus a fixed approach margin.

E. Energy-Based Candidate Scoring, as an Alternative to OT-CFM

Having found that OT coupling specifically degrades candidate generation (section IV), we tested a different paradigm rather than only patching the flow-matching training recipe: an implicit, energy-based model (EBM) that scores candidate poses directly, motivated by Implicit Behavioral Cloning [5] and its grasp-specific descendant [6], both of which report that implicit/EBM policies are more sample-efficient than explicit generators on small, multimodal, discontinuous action distributions – exactly our regime (~ 400 examples/object).

Why an EBM sidesteps the OT-coupling failure.

An explicit generator (CFM, diffusion) must learn a full noise-to-data *trajectory* that is well-behaved across the whole conditioning set; minibatch OT coupling is one way of regularizing that trajectory, and section IV-D shows it backfires under our small per-condition data budget. An EBM instead only needs to learn a scalar compatibility score $E_{\theta}(\text{pose}, c) \in \mathbb{R}$ from contrastive success/failure labels – no noise-to-data pairing of any kind is involved, so there is no coupling mechanism to get wrong.

Model and training data. E_{θ} is a 4-layer MLP scoring $(x, y, \sin \psi, \cos \psi, c) \in \mathbb{R}^{260}$ (the same meaningful pose dimensions and SAM conditioning as OT-CFM). Training reuses the *same* labeled dataset LGGSN and OT-CFM

both already train on: 2,853 label=1 (physically successful) rows as positives and 4,147 label=0 (failed) rows as one negative source – no new data collection.

A documented failure mode, and its fix. A first version trained with plain binary cross-entropy against the static positive/negative labels, then sampled at inference via cross-entropy-method (CEM) search over (x, y, ψ) guided by E_θ . This failed catastrophically (0–16% physical success across three objects tested, section IV-D) – markedly worse than every other method we tried. Direct inspection showed why: CEM converged to a normalised pose ~ 1.7 standard deviations from the training mean, where the model assigned an artificially high score (99% confident) despite this joint region being essentially unconstrained by training data – each marginal dimension looked plausible in isolation, but the specific combination was never taught to be low-scoring. This is precisely the failure Implicit Behavioral Cloning’s training recipe is designed to prevent, and our naive version omitted the relevant piece: IBC does not train against a fixed negative set, but against negatives *mined from the model’s own current beliefs during training*, so any region the model starts to over-score is explicitly corrected before training ends.

We re-trained E_θ with this mechanism: an InfoNCE-style loss (softmax cross-entropy over {positive, negatives} with the positive as the target class) against a negative pool assembled, per positive example, from three sources – (i) 4 *static* negatives (real logged label=0 rows for the same object, free), (ii) 4 *uniform* negatives (sampled across the training data’s observed pose range, for broad coverage), and (iii) 4 *hard* negatives *mined via a short 3-iteration CEM search using the model’s own current weights*, shared across that object’s positives within the current minibatch. This last term is the one that was missing and is what prevents the inference-time CEM search from later exploiting an under-constrained region: whatever the model currently over-scores becomes a training negative for that very step. **Inference.** Identical CEM search as the naive version (population 64, 6 iterations, elite fraction 0.2), unchanged – only the training procedure differs between the two EBM variants reported in section IV-D, isolating the effect of adversarial negative mining from any change to the sampling algorithm itself.

F. LGGSN Reranker

We use the LGGSN v5d reranker [14] without modification. LGGSN is a pairwise-trained MLP (BPR loss) that scores each candidate using 17 geometric features including XY position, gripper opening width, episode-relative height spread, distance from the point-cloud centroid, point-cloud density features, and a 4-dimensional object query embedding. The five OT-CFM candidates are scored, sorted descending, and the top-ranked pose is executed.

IV. EXPERIMENTS

A. Evaluation Protocol

All experiments run on MuJoCo/SO-ARM101 with physics-based bilateral contact verification as the grasp

TABLE I
OVERALL SUCCESS RATE: 350 TRIALS (7 OBJECTS $\times$ 50 SEEDS),
FIXED-SEEDING HARNESS.

Method	SR	95% CI	Δ pp	z	p
Baseline (random CoM + LGGSN)	79.1%	[74.6, 83.1]	—	—	—
OT-CFM + LGGSN	69.1%	[64.1, 73.8]	−10.0	−3.02	0.0025**

success criterion. We evaluate on 7 YCB objects (Banana, TomatoSoupCan, Pear, MustardBottle, Scissors, CrackerBox, PowerDrill) over 50 random seeds each, giving **350 trials per condition**. Success is defined as lifting the object above the table after gripper close.

Baseline. Random CoM sampling ($x \sim g_{\text{com}} \pm 0.06$ m, $\psi \sim \mathcal{U}[-\pi/2, \pi/2]$, $w \sim \mathcal{U}[0.04, 0.09]$ m, same LGGSN v5d reranker) produces 5 candidates per episode.

A confirmed evaluation bug, and how we responded to it. An earlier version of this evaluation (175 trials, 25 seeds/object) never passed a per-trial seed into the generator’s inference-time sampling call. Because that call imports `train_cfm_grasp`, which set `torch.manual_seed(42)` at *module* level, every trial in a freshly started process drew noise from whatever state that fixed seed left behind at import time, independent of the trial’s nominal `-seed` – so 25 nominally distinct trials per object executed close to the same candidate. Random-CoM sampling was unaffected (it draws from `numpy`’s properly-seeded generator, not the generator network), which is why, in retrospect, only the learned-generator conditions showed the suspiciously narrow, high success rates reported in an earlier draft of this paper. We moved the seed into the sampling call itself, re-ran every condition at $n=50$ /object under the fixed code, and report only the corrected numbers throughout this paper; section IV and section IV-D are a direct account of what changed and why.

Statistics. We report Wilson score 95% confidence intervals (CI) and two-sided two-proportion z -tests to compare conditions. Significance thresholds: ** $p < 0.05$, * $p < 0.10$.

B. Main Result

table I reports the primary comparison, on the corrected, per-trial-seeded harness. OT-CFM + LGGSN achieves 69.1% (242/350) versus the baseline’s 79.1% (277/350) – a statistically significant *decrease* of -10.0 pp ($z = -3.02$, $p = 0.0025$). This is the opposite conclusion from the $+12.0$ pp gain reported under the unfixed, unseeded harness, and the 95% Wilson confidence intervals are disjoint in the opposite direction: the learned generator’s candidates are reliably *worse*, not better, than uniform random sampling around the object’s center of mass, once the generator is actually resampled per trial instead of executing a near-constant candidate.

section IV-D isolates why: the failure is specific to OT coupling, not to learned candidate generation in general, and section III-E reports an alternative (energy-based scoring, properly trained) that reaches parity with this baseline rather than falling significantly below it.

TABLE II

PER-OBJECT SUCCESS RATE: BASELINE VS. OT-CFM VS. EBM v2 (50 SEEDS EACH). EBM v2 EXCLUDES SCISSORS (§III-E: ITS OBJECT NAME NEVER MATCHES THIS CHECKPOINT’S TRAINED CONDITIONING KEYS, AN UNRELATED BUG THAT SILENTLY FALLS BACK TO BASELINE FOR THAT OBJECT, EXACTLY AS IT DOES FOR OT-CFM IN SECTION V-C).

Object	Base	OT-CFM	Δ	p	EBM v2	Δ	p
Banana	100.0%	96.0%	-4.0	0.153	100.0%	0.0	—
TomatoSoupCan	92.0%	78.0%	-14.0	0.050**	100.0%	+8.0	0.041**
Pear	72.0%	52.0%	-20.0	0.039**	80.0%	+8.0	0.349
MustardBottle	84.0%	70.0%	-14.0	0.096*	70.0%	-14.0	0.096*
Scissors	88.0%	88.0%	0.0	—	—	—	—
CrackerBox	42.0%	28.0%	-14.0	0.142	26.0%	-16.0	0.091*
PowerDrill	76.0%	72.0%	-4.0	0.648	68.0%	-8.0	0.373
All (7)	79.1%	69.1%	-10.0	0.0025**	—	—	—
All (6, excl. Scissors)	77.7%	66.0%	-11.7	0.0015**	74.0%	-3.7	0.294

C. Per-Object Analysis

table II shows every object’s OT-CFM Δ is negative or zero, and 3 of 7 individually reach significance ($p < 0.05$) in the *harmful* direction (TomatoSoupCan, Pear) or trend that way (MustardBottle, CrackerBox, $p < 0.15$). Only Banana and PowerDrill show small, non-significant drops. EBM v2 is more mixed: it significantly *beats* baseline on TomatoSoupCan, is directionally positive on Pear, and is directionally negative (not significant) on MustardBottle, CrackerBox, and PowerDrill—consistent with the pooled, non-significant -3.7 pp. No per-object story we tried (OT-CFM’s specific failure excepted) reaches a clean, uniform verdict across all seven objects; we report the full spread rather than the two or three objects that would look best in isolation.

Landing-position sensitivity, independent of method. A separate, generation-isolated diagnostic¹ asks a narrower question than table II: for a *fixed* sampler, does exactly where a candidate lands on the object predict success? This is answered independently of whether any given generator is good at finding those positions. MustardBottle and Pear show the strongest, Bonferroni-surviving signal (local point density/normal consistency, $p=0.00030/0.00029$ and $p=0.00068$); TomatoSoupCan survives Bonferroni on local point density ($p=0.00123$); CrackerBox and TomatoSoupCan show weaker, BH-only signal on the remaining features; PowerDrill is the only object with a clean null across all three features under either correction ($p \geq 0.17$). A post-hoc follow-up (single exploratory test, not part of the pre-specified 15-test family, uncorrected) checks a fourth, elongation-specific feature for PowerDrill—gripper-axis alignment relative to the object’s principal axis—and finds only a marginal trend ($p=0.062$): PowerDrill’s point cloud is only weakly elongated (major/minor eigenvalue ratio 1.075), so PCA’s

¹Single-object scenes, semantic grounding disabled, 50 OT-CFM trials/object (a different harness and seed grid from table II, collected before the seeding-bug audit on the already-per-trial-seeded worktree branch that was later merged—see section IV—so this diagnostic’s own candidates were never affected by the bug). Three contact features (local point density, surface-normal consistency, contact-width ratio) are tested success- vs. fail-trial with Mann-Whitney U , corrected across 3 features $\times$ 5 objects (Pear, MustardBottle, CrackerBox, TomatoSoupCan, PowerDrill) via Bonferroni and Benjamini-Hochberg.

TABLE III

DIAGNOSTIC COMPARISON ON THE THREE OBJECTS WHERE OT-CFM’S FAILURE IS CLEAREST. BASELINE/OT-CFM/EBM v2 AT THEIR FULL $n=50$; THE FOUR DIAGNOSTIC CONDITIONS AT $n=25$ (SAME FIXED-SEEDING HARNESS). p VS. BASELINE.

Condition	Pear	TomatoSoupCan	CrackerBox
Baseline ($n=50$)	72%	92%	42%
OT-CFM ($n=50$)	52%**	78%**	28%
Remove-OT (no coupling)	72%	100%	28%
DDPM (ε -pred., DDIM-50) [‡]	68%	100%	24%
Stratified-OT (C2OT fix, §III-C)	52%	88%	44%
EBM v1 (naive BCE training)	0%**	0%**	16%**
EBM v2 (mined negatives, $n=50$)	80%	100%**	26%*

“principal axis” is not a strongly stable direction here, and PowerDrill’s null remains the most plausible reading. We caution against over-interpreting this diagnostic as explaining table II’s Δ column: it establishes that landing position matters (a lot, for some objects), not that any particular generator we tested reliably finds the better positions—the main table shows OT-CFM specifically does not, and EBM v2 does only for TomatoSoupCan.

Scissors (88%, both conditions). Scissors recovers from 0/25 to 22/25 under both methods after correcting a physics-configuration bug: the original VHACD collision mesh (14 merged convex hulls, ≈ 2 cm thick) tunnelled through the table under MuJoCo contact detection. The corrected entry uses a bounding-box collision proxy scaled to 4 cm thickness—only marginally above the gripper’s 4 cm minimum opening, which we found makes this specific object’s success rate sensitive to run-to-run variation: independent 25-seed re-measurements of the identical configuration on three separate dates gave 25/25, 23/25, and 22/25 (section V-C discusses this instability; we report the most recent measurement here rather than treat any one run as authoritative). Since baseline and OT-CFM are numerically identical for Scissors, it contributes equally to each condition and does not differentiate methods.

D. Ablation Study

** $p < 0.05$, * $p < 0.10$ vs. Baseline. [‡]Same MLP architecture and training data as OT-CFM; cosine schedule $T=1000$.

table III decomposes *why* OT-CFM fails and what, among the alternatives we tried, actually recovers baseline-competitive performance.

The failure is specific to OT coupling, not to learned candidate generation. Remove-OT (identical architecture and training data, no minibatch OT coupling) exactly matches baseline on Pear (72% vs. 72%, $z=0.00$) and is directionally at or above baseline on TomatoSoupCan (100% vs. 92%, $p=0.15$); only CrackerBox stays below baseline (28% vs. 42%, $p=0.24$, not significant). DDPM (same architecture and data, diffusion instead of flow matching) shows the same pattern. Both controls change one thing relative to OT-CFM – whether the training pairs are OT-coupled – and both track baseline far more closely than OT-CFM does. This localizes the problem to the coupling mechanism itself, consistent with Cheng

and Schwing [4]: minibatch OT coupling that ignores the conditioning variable creates a training-time prior that inference-time sampling, drawing from the full unconditional prior, was never trained to match. Our training minibatches mix all 7 object conditions, so this is exactly the failure mode their analysis predicts.

A condition-aware fix helps partially and inconsistently, not cleanly. We adapted Cheng and Schwing’s [4] discrete-class fix to our 7-object setting: rather than a shared OT plan across the mixed minibatch, we solve OT coupling independently within each object’s own training examples (the discrete-class limit of their released configuration, which uses a very large condition weight to the same effect). This recovers TomatoSoupCan partway (78%→88%, still below baseline’s 92%) and helps CrackerBox (28%→44%, now above baseline, though CrackerBox is noisy for every method tested), but leaves Pear completely unmoved (52%, identical to unfixed OT-CFM) – the object where OT-CFM’s failure is largest. We do not read this as the fix being wrong in principle, but as insufficient on its own in our data regime: Pear has the fewest training examples of the three objects shown here, and a per-object OT sub-batch of that size may simply be too small for coupling, condition-aware or not, to help. We report this as a partial, honest result rather than omit it.

A naive energy-based model fails catastrophically, and we show exactly why. EBM v1, trained with plain binary cross-entropy against static labels and sampled via CEM search, collapses to 0% on two of three objects ($p<0.001$ both) – worse than every other condition tested, including OT-CFM. section III-E traces this to CEM exploiting a region the model never learned to score correctly, and describes the fix: training with hard negatives mined from the model’s own current beliefs, as Implicit Behavioral Cloning [5] prescribes and our first attempt omitted.

The corrected EBM (v2) reaches baseline parity, with one individually significant win. With adversarial negative mining added and nothing else changed, the same architecture recovers to 80%/100%/26% – matching or exceeding baseline on Pear and TomatoSoupCan (significantly so on the latter, $p=0.041$), and remaining below baseline only on CrackerBox, the object every generative method we tested struggles with. Pooled across all six objects with EBM v2 coverage (table II), the difference from baseline is not significant (−3.7 pp, $p=0.29$) – in contrast to OT-CFM’s significant, uniformly negative pooled result. We read this as evidence that properly-trained implicit scoring is a viable, if not clearly superior, alternative to explicit generative candidate proposal in this small-data regime, not as a second method that beats baseline the way OT-CFM was originally (incorrectly) reported to.

Reliability across generation seeds. table III compares methods at one generated candidate set per orientation seed (25 orientation seeds). To check whether the OT-CFM-vs.-alternatives ranking holds up under independently resampled candidates, we ran a separate, generation-isolated experiment (semantic grounding disabled, single-

object scenes, 5 orientation seeds $\times$ 10 generation seeds per object, the same OT-CFM/CFM-noOT/DDPM checkpoints). Pairing trials by (object, orientation seed, generation seed) and testing with McNemar’s exact test, OT-CFM is significantly *less* reliable than both alternatives specifically on Pear ($p=0.041$ vs. CFM-noOT, $p=0.021$ vs. DDPM) and TomatoSoupCan ($p=0.012$, $p=0.002$), with no significant difference on the remaining objects. We report this as a limitation rather than a contradiction of table III: the two experiments differ in evaluation harness (semantic grounding on vs. off) and seed structure and are not directly comparable, but the seed-resampled result indicates OT-CFM’s per-object advantage is not uniform across generator draws and should not be assumed to generalize to every resampling of the generator. An ensemble candidate-selection strategy substantially mitigates this for both flagged objects: drawing 10 independent OT-CFM candidates (same generation-isolated harness) and executing the one closest to their pose-space consensus (median), rather than the one with the lowest predicted IK error, raises Pear’s success rate from 6% to 68% (Fisher’s exact $p=5.8\times 10^{-11}$) and TomatoSoupCan’s from 34% to 64% ($p=0.0048$), at matched ensemble size (10 vs. 10, so the effect is attributable to the selection rule and not pool size). MustardBottle and CrackerBox—which show no significant seed-reliability gap above—show a correspondingly weaker (MustardBottle, 50%→68%, $p=0.10$) or absent (CrackerBox, 44% vs. 44%, exact tie) benefit from the same strategy, consistent with the mitigation targeting specifically the objects where single-draw reliability is worst.

V. DISCUSSION

A. Why OT Coupling Hurts Here

Minibatch OT coupling [2] was designed for and validated on *unconditional* generation, where solving one Earth-Mover assignment per minibatch is unambiguous: every noise sample is interchangeable with every data sample. Our setting is conditional – each minibatch mixes seven object identities – and Cheng and Schwing [4] show this changes the picture: a shared, condition-agnostic OT plan produces a training-time prior over x_0 that is implicitly skewed by which noise samples the solver happened to assign to which object in that minibatch, while inference-time sampling draws x_0 from the full, unconditional $\mathcal{N}(0, I)$ prior the model never actually trained against for a given condition. section IV-D’s controls are consistent with exactly this: removing OT coupling (Remove-OT) or replacing flow matching with diffusion (DDPM) – changing one thing relative to OT-CFM – both track baseline far more closely than OT-CFM does, and a discrete-class-appropriate condition-aware fix (Stratified-OT) partially recovers two of three objects tested. None of this would follow if the flow-matching paradigm itself, or the training data, were the problem; it is specific to the coupling mechanism, at our per-condition data scale (~ 400 examples/object). We flag this as a caution for the growing body of robotics work that adopts minibatch-OT-coupled flow matching by

default, citing its unconditional benefits, without checking whether their training minibatches mix conditions the way ours do.

B. Why a Properly-Trained Energy-Based Model Reaches Parity

section III-E and section IV-D report a second failure-and-fix pair, structurally similar to the first but at a different layer of the pipeline: a naive EBM trained with static contrastive labels is exploited by its own inference-time search into a confidently-wrong region the training loss never constrained; adding hard-negative mining – training against whatever the model currently over-scores, not only the original dataset’s logged failures – fixes this, exactly as Florence et al. [5] describe and Singh et al. [6] apply to grasp generation specifically. Both failure modes we report (condition-blind OT coupling, un-mined EBM negatives) share a common shape: a training procedure that is correct on average but leaves a specific, structured blind spot that the deployed inference procedure (unconditional-prior sampling; adversarial CEM search) reliably finds and exploits. Neither is a criticism of flow matching or energy-based models as paradigms; both are reminders that the training recipe and the inference procedure must be checked jointly, not validated separately.

C. Scissors: Physics-Configuration Correction

Scissors was 0/25 across all conditions due to a VHACD mesh tunnelling bug: 14 merged convex hulls at ≈ 2 cm thickness allowed the object to pass through the table under MuJoCo contact detection. Replacing the geometry with a bounding-box proxy at 4 cm restores a high success rate for all methods, but not a clean ceiling: the proxy’s 4 cm thickness is only marginally above the gripper’s 4 cm minimum opening, so grasp success is sensitive to run-to-run variation in exactly where the gripper contacts the (barely-cleared) collision geometry.

We measured this configuration on three separate dates and got three different counts out of 25 seeds: 25/25 (2026-06-26), 23/25 (2026-06-28, reported in project history but without a surviving per-seed log), and 22/25 (2026-07-09, the measurement reported in this paper, verified with a full per-seed log and a smoke test confirming the executed candidates and outcomes vary meaningfully across seeds, i.e. this is not a degenerate or stuck sampler). We do not have an explanation for the drift—the manifest-level physics configuration is unchanged across all three dates—beyond the geometry’s thin safety margin making it more susceptible than other objects to small numerical differences between runs (e.g. contact-solver iteration order, floating-point accumulation). We report the most recent (2026-07-09) measurement rather than treat any single run as ground truth, and flag this instability as a property of this specific object’s collision geometry, not of the grasping pipeline being evaluated: baseline and OT-CFM are affected identically (22/25 for both, at every re-measurement above), so Scissors still contributes equally to each condition and

does not differentiate methods, regardless of which of the three counts is used.

A second, unrelated issue also affects Scissors in a separate, supplementary generation-isolated diagnostic experiment (not reported in this paper): the object-name matching used to condition OT-CFM sampling never matches “scissors” against its trained category key, silently falling back to random sampling in that other experiment. This does not affect any number reported here—baseline and OT-CFM are identical for Scissors in this paper’s own evaluation regardless of which sampler produced the candidates—but we note it for transparency since both issues share the same object.

D. Limitations

Top-down constraint. The current pipeline fixes roll $= \pi$, pitch $= 0$, reducing the 6-DoF problem to effective 3-DoF $((x, y, \psi))$. Extension to arbitrary gripper orientations requires either SE(3)-equivariant flow matching or a richer orientation parameterisation (e.g., quaternion or 6-d representation).

Simulation only. All results are from MuJoCo simulation. Sim-to-real transfer requires alignment of the physical SO-ARM101’s calibration, contact dynamics, and depth-camera intrinsics with the simulation geometry.

CoM-relative shift. eq. (5) centres generated candidates on the current object CoM by subtracting the training-set global mean. This assumes the training object-placement distribution is roughly centred, which holds for our setup but may break for out-of-distribution workspaces.

No generative method we tested reliably beats baseline on CrackerBox. Baseline, OT-CFM, Remove-OT, DDPM, Stratified-OT, and EBM v2 are all at or below baseline on CrackerBox (table III), the one object where the pattern seen everywhere else – some method reaching parity or better – does not appear. We do not have a settled explanation; §IV-C’s contact-feature diagnostic finds only a weak, BH-only signal there, consistent with CrackerBox’s flat geometry being intrinsically harder for any of the position/orientation-based methods tested, but this is not confirmed.

Not fully swept; one check finds Pear specifically fragile. Stratified-OT’s failure to move Pear at all, unlike its partial recovery of TomatoSoupCan/CrackerBox, may reflect Pear’s smaller training set (248 positives, fewest of all 7) rather than a limit of the approach. A stronger EBM v2 hard-negative-mining schedule reproduces TomatoSoupCan/ CrackerBox exactly (100%/28%) but drops Pear 72%→48% (ns at $n=25$, McNemar $p=0.15$) – again Pear, again the least data.

E. Future Work

The clearest next step is closing the gap this paper leaves open on Pear and CrackerBox: whether Stratified-OT’s partial recovery is a data-budget limit (testable by rebalancing or augmenting Pear’s training set) or a deeper limitation of the discrete-class coupling fix, and

whether a wider hard-negative-mining budget or a learned (rather than uniform-random) proposal distribution for CEM search closes the remaining EBM v2 gap on Cracker-Box. SE(3)-equivariant flow models (e.g., Riemannian flow matching [9]) or an SE(3)-aware energy function could extend either approach to the full 6-DoF pose space, relaxing the top-down constraint (section III-D); validating on a Piper 6-DoF manipulator would test this directly. Real-robot evaluation on the physical SO-ARM101 is a further validation step, independent of which candidate-generation method is used. More broadly, we encourage other groups applying minibatch-OT-coupled flow matching to conditional, small-per-class-data robotics problems to check for the failure mode in section V-A before assuming OT coupling’s unconditional-generation benefits transfer.

VI. CONCLUSION

We set out to build a grasp generation pipeline combining Optimal Transport Conditional Flow Matching (OT-CFM) with geometric reranking (LGGSN), and an initial 175-trial evaluation appeared to show a large, significant gain over random-CoM sampling. Auditing the evaluation harness before finalizing this result uncovered a genuine bug – unseeded generator sampling silently collapsed 25 nominally distinct trials per object to near-repeats of one candidate – and fixing it reversed the conclusion: on a clean, per-trial-seeded $n=50$ /object re-evaluation (350 trials/condition), OT-CFM is significantly *worse* than random-CoM sampling with the same reranker (-10.0 pp, $z = -3.02$, $p=0.0025$).

We used the corrected pipeline as a diagnostic instrument rather than discarding it. Controlled comparisons localize the failure to minibatch optimal-transport coupling specifically, not to conditional flow matching or learned candidate generation in general: removing OT coupling, or replacing flow matching with diffusion, both track baseline far more closely than OT-CFM does, consistent with a training/inference distribution mismatch recently identified for conditional flow matching [4] – to our knowledge the first demonstration that this mismatch degrades physically executed robot task success, not only a generation-quality proxy. A condition-aware coupling fix recovers two of three objects tested partially, but not Pear, and not cleanly.

We separately tested an energy-based alternative motivated by Implicit Behavioral Cloning [5]. A naive version, trained without adversarial negative mining, was exploited by its own inference-time search into a confidently-wrong, data-unsupported region (0–16% success); adding the negative-mining mechanism IBC prescribes and our first attempt omitted recovers the same architecture to statistical parity with baseline (74.0% vs. 77.7%, $p=0.29$), significantly exceeding it on one of six objects. Across every generative candidate method we tested – OT-CFM, OT-free CFM, DDPM, a condition-aware OT variant, and two energy-based variants – none reliably and significantly beats a well-reranked random baseline in this small-data (~ 400 examples/object), single-GPU regime; the one significant effect in either direction is OT-CFM’s specific, avoidable

harm. One method does help robustly and without any retraining: selecting the pose-space consensus candidate from ten independent draws, rather than the lowest-predicted-IK-error one, significantly improves reliability on exactly the two objects where single-draw sampling is least stable. We report all of this, including what did not work and why, because we believe the diagnosis and the two documented failure-and-fix pairs are of independent value to others building similar pipelines under similar data and compute constraints.

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